Plan of Action: Implementing the Veeva Integrated Curriculum Program for the 2027 Batch

To successfully conduct the Associate Automation Engineer curriculum for the **Veeva Integrated Curriculum Program** for college graduates, we define a structured approach that bridges **foundational software testing** with **specialized industry applications.**

Curriculum Start Date: October 19 2026

Week-by-Week Plan of Action

Week 1: Testing Foundation and Core Java (19-26 October 2026)

Focus: The focus is to validate understanding of software testing fundamentals—including test levels, design, and lifecycles—alongside the practical implementation of core Java concepts like OOPs, collections, exception handling, and automation framework development.

Topics Covered:

SDLC

❖ Requirements Design

❖ Coding

❖ Testing

❖ Deployment

Fundamentals of Testing

❖ Quality Assurance

❖ System Integration Testing

❖ User Acceptance Testing

❖ Agile vs Waterfall

❖ Bug Tracking Tool (JIRA) usage

❖ Test Case Management (XRay) usage

❖ Confluence usage

Hands-on Practical Skills:

➢ Set up a development environment using Intellij IDE.

➢ Basic Java programming hands-on leveraging the concepts mentioned

- **Practical Skills for Bug Tracking in Jira**
  - **Project and Board Setup**
  - **Writing Effective Bug Tickets**
  - **Triage, Assignment, and Linking**
  - **Advanced Searching and Filtering (JQL)**
  - **Jira Query Language (JQL)**
  - **Custom Views & Dashboards**

➢ Understand test life cycle using XRay - create and execute tests

- **Test Design (Creating Artifacts)**
  - **Creating a Test Case**
  - **Adding pre-conditions**
  - **Establishing Traceability**
- **Test Organization**
  - **The Test Repository (Hierarchical)**
  - **Test Sets (Flat Lists)**
- **Test Execution**
  - **Setting up a Test Plan**
  - **Creating a Test Execution**
- **Running the Test**
- **Reporting & Dashboard Metrics**
  - **Requirement Coverage**
  - **Traceability Report**
  - **Test Plan Metrics**

Week 2: Core Java Concepts (27 October-4 November 2026)

Focus: Focus on mastering Object-Oriented Programming (OOP) principles, robust memory management, Exception Handling, and the Collections Framework to build scalable and efficient applications

Topics Covered:

Basics

❖ Class, Object and package

❖ Data Types

❖ Control Flow

❖ Methods

❖ Code debugging

❖ Loops, Arrays

OOPS

❖ Encapsulation

❖ Inheritance

❖ Polymorphism

❖ Abstraction

Collections

❖ Lists

❖ Set

❖ Map

Streams

❖ Intermediate operations

❖ Terminal operations

Lambda Expressions

❖ Syntax & use cases

Exception Handling

❖ try, catch, finally

❖ Throw/ Throws

❖ Custom exceptions

Java I/O Operations

❖ Read and Write File

❖ Creation of File

❖ Buffer Reader

❖ Buffer Writer

❖ File Permissions

Hands-on Practical Skills:

➢ Apply OOPS principles to build simple, modular Java classes.

➢ Basic Java programming hands-on leveraging the concepts mentioned

➢ Use appropriate collections for different scenarios.

➢ Perform data manipulation with Streams.

➢ Implement robust error handling

➢ Write concise, functional code using Collections and Lambdas.

➢ Write code to perform CRUD operations on files.

Week 3: **Advanced Java - Design Patterns and Introduction to Test Automation**

(05 -12 November 2026)

Focus: The Focus is to provide reusable, proven blueprints for writing scalable, flexible, maintainable, and loosely coupled enterprise software

Topics Covered:

Design Patterns

- ❖ Singleton
- ❖ Builder Pattern

Automation Fundamentals & Environment Setup

- ❖ **Why automation? Manual vs. Automated testing.**
- ❖ **Selenium suite overview (IDE, WebDriver, Grid).**

Hands-on Practical Skills-Design:

➢ Build a simple test automation configuration manager.

➢ Use Singleton for ConfigurationManager

➢ Use Builder for TestConfig

➢ Read configuration values such as browser, URL, timeout etc.

➢ Create multiple TestConfig objects using Builder.

➢ Ensure only one ConfigurationManager instance exists

Hands-on Practical Skills- Selinium Introduction

- ➢ Environment setup: Installing Java/JDK, Eclipse/IntelliJ, and configuring environment variables.
- ➢ Creating the first Selenium script (Launch browsers: Chrome, Firefox, Edge).

Week 4: Build Tools - Maven and **Test Automation** - Selenium

(13 -20 November 2026)

Topics Covered:

Maven

❖ Project structure

❖ Maven lifecycle

❖ Goals

❖ Scope of each Goal

❖ Dependency Management

❖ Plugins configuration

❖ Settings.xml file

❖ Profiles

❖ Nexus Repository config

Selenium

❖ Webdriver setup. Browser initialization

❖ Locating elements (DOM structure)

❖ CSS Selectors

❖ Webdriver Commands ➢ Navigation

➢ Element interaction

❖ Dropdown Actions

❖ Screenshot Capture

❖ Windows and Frame Handling

❖ Multi Tab handling

❖ Page Object Model (POM)

- ❖ Synchronization
    - ➢ Implicit vs Explicit vs Fluent waits
- ❖ Verification & Validation Actions
- ❖ Mouse & Keyboard Actions

Hands-on Practical Skills-Maven:

➢ Create multi-module Maven projects managing project dependencies

➢ Create different maven profiles

➢ Configure surefire, compiler, fail safe, etc. plugins in pom.xml file

➢ Setup local nexus repository

➢ Configure Settings.xml file to access the artifacts from local nexus repository

Hands-on Practical Skills-Selenium:

➢ Browser initialization and locate DOM elements

➢ Write robust web element locators.

➢ Perform basic user actions on a web page

➢ Implement effective waits to handle dynamic pages

➢ Develop reliable test steps

➢ Create a complete automation framework (Page Object Model) using all learned tools.

Assessment-1 : (21-25 November 2026)

Week 5: **Version Control -** Git Operations, Playwright and REST API Basics

(21-30 November 2026)

Topics Covered:

Git Operations

❖ Repo cloning via ssh

❖ Repo cloning via https

❖ Difference ssh vs https clone

❖ Clone Repo

❖ Fetch and Pull

❖ Commit and Push

❖ Branching

❖ Merge and Rebase

❖ Commit diff

Playwright

❖ What is a Playwright?

❖ Playwright vs Selenium

❖ Install Playwright

❖ Playwright supported tech stack

❖ Cross Brower testing

❖ Handing Web Elements.

❖ Waits & Synchronization

REST API Basics

❖ REST Architecture

❖ REST API Principles

❖ Status codes

Hands-on Practical Skills-Maven

➢ Create a local git repository and push it to remote

➢ Create a branch using git command line and push it remote

➢ Rebase a local branch with remote

➢ Commit message using git command line

➢ Pull latest code from remote branch and fix merge conflicts

Hands-on Practical Skills- Playwright, REST API Basics

➢ Setup node.js and Playwright in local machine

➢ Create a playwright project

➢ Understand project folder structure

➢ Understand playwright.config

➢ Run first UI playwright test

➢ Run the tests from command line

Week-6: **Testing Frameworks** – TestNG **and Linux - Linux basics**

Topics Covered

TestNG

❖ Annotations

❖ Grouping

❖ Parameterization

❖ Data Providers

❖ Listeners

❖ Assertions

- ➢ Soft
- ➢ Hard

Linux

❖ Basic Linux Commands

❖ Linux File System

❖ File Management in Linux

❖ File Permissions

Hands-on Practical Skills- TestNG

➢ Convert existing Selenium tests to TestNG

➢ Read the data from csv file in Data provider

➢ Create a group of regression Test cases and run it

➢ Debug test case failures

Hands-on Practical Skills-Linux

➢ Find a file in Linux using a command

➢ Change file permission

➢ Check CPU memory and disk usage

- ➢ Read logs and understand the root cause
- ➢ Create and extract .tar.gz files
- ➢ grep a term (word) and list the files that has the word occurrences
- ➢ Create and execute basic shell script

Week-7: Postman and API Automation and **BDD** - Gherkin – Cucumber **(1-8 December 2026)**

Topics Covered

Postman

❖ Request building ➢ GET

➢ POST

➢ PUT

➢ DELETE

❖ Environment variables

❖ Basic scripting (pre-request/ Tests)

API Automation

❖ REST Assured

- ➢ Setup
- ➢ Building requests (params, headers, body)

❖ Authentication

- ➢ Basic/ Bearer token

❖ Response validation

- ➢ JSON/ XML parsing
- ➢ Schema validation

Gherkin

❖ Given

❖ When

❖ Then

❖ And

❖ *

Cucumber

❖ Feature files

❖ Scenario

❖ Scenario Outline

❖ Background

❖ Rule

❖ Step definitions

❖ Runner File

❖ Tags

❖ Cucumber Options

❖ Hooks

Hands-on Practical Skills- Postman

➢ Manually test API using Postman

➢ Write basic Postman test assertions

➢ Automate CRUD operations with REST Assured

➢ Integrate API tests into the Maven build.

Hands-on Practical Skills- API Automation

➢ Create a post api call with basic authentication

➢ Read the Post call body from the Json file.

➢ Perform GET API schema validation against the response

Week-8: **DevOps** - DevOps Basics – Jenkins and **Cloud Fundamentals** - AWS

(09 -18 December 2026)

Topics Covered

DevOps Basics

❖ Phases of DevOps lifecycle

Jenkins

❖ Setup Jenkins in local machine

❖ Setting up CI/CD pipelines

❖ Cron Schedules

❖ Pipeline stages

❖ Parallel stages

❖ Stage dependency

❖ Trigger types

❖ SCM polling

❖ Webhooks

❖ Debugging pipeline

❖ Failures & Common issues

❖ Log analysis

AWS

❖ Accessing AWS

❖ Management console

❖ Command line interface (CLI)

❖ S3, EC2, EBS, AMI, Lambda

❖ Other elastic computes (ECS, EKS, ECR)

Hands-on Practical Skills- BDD - Gherkin – Cucumber

➢ Implement BDD feature files using Cucumber.

➢ Write reusable step definitions

➢ Write different test cases using Scenario and Scenario outline

➢ Create cucumber hooks class

➢ Configure cucumber options in runner class

Hands-on Practical Skills- DevOps

➢ Install Jenkins locally

➢ Configure Java/JDK

➢ Access Jenkins dashboard

➢ Create and manage users

➢ Configure basic Jenkins settings

➢ Create a Freestyle project

➢ Configure source code repository

➢ Configure build steps

➢ Configure post-build actions

➢ Run jobs manually

➢ Analyze and fix a failed test case.

Hands on Practical Skills: AWC-Cloud

➢ Register AWS account

➢ Create EC2 machines for Linux and Windows.

➢ Create S3 buckets

➢ Store data in S3 bucket

➢ Configure object permissions

➢ Enable bucket versioning

➢ Create an EBS volume

➢ Attach an EBS volume to EC2

➢ Detach and reattach an EBS volume

➢ Understand EBS availability-zone dependency

Assessment-2 (21-25 December 2026)

Week-9: and **AI Fundamentals** - Gemini - Claude **(19 -31 December 2026)**

Gemini Web

❖ Basics

❖ Writing good Prompts

❖ Creating GEMS

Claude

❖ Basics

❖ Writing good Prompts

❖ Using Claude CLI

❖ Creating md files, skills

❖ Creating agents and MCPs

❖ Integration with Playwright

Hands on Practical Skills: **Gemini and Claude**

➢ Create a Gemini GEM for repetitive tasks.

➢ Create a Claude skill for repetitive tasks using CLI

➢ Create, read, and edit md files with Claude

➢ Generate and refactor automation code using Claude

➢ Generate framework documentation using md file

➢ Connect Claude to an MCP server

➢ Build a custom MCP server using Claude API

Week-10,Week-11 , Week-12 and Week-13: Project Work

(1 January to 5 February 2027)

Assessment-3 : (21-25 January 2027)